

OMAR HAZEM

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Profile

Computer Science student with strong interests in Artificial Intelligence, Machine Learning, Deep Learning, and Large Language Models (LLMs). Experienced in Python development, computer vision, and building AI-based projects using modern deep learning frameworks. Passionate about research, problem solving, and developing scalable AI systems while continuously improving software engineering and deployment skills.

Education

Zewail City for Science and Technology

2023 – Expected 2027

B.Sc. in Data Science, Artificial Intelligence

Experience

Digital Egypt Pioneers Initiative (DEPI) - MCIT

Oct 2024– Jun 2025

IBM Data Scientist Intern

Cairo, Egypt

- Participated in a national-level Data Science and Artificial Intelligence internship supported by IBM and MCIT.
- Built and evaluated machine learning models using real-world datasets and industry-oriented workflows.
- Worked with supervised learning algorithms including Random Forest, XGBoost, Support Vector Machines, Decision Trees, and Neural Networks.
- Applied hyperparameter tuning, feature engineering, and cross-validation to improve model performance.
- Developed end-to-end machine learning pipelines using Python, Pandas, scikit-learn, and Jupyter Notebooks.
- Completed 30+ practical projects covering data preprocessing, exploratory data analysis (EDA), model development, and evaluation.
- Documented technical work and collaborated with mentors and peers in team-based assignments and live sessions.

Mostaqel

Apr 2025– Sep 2025

Freelance Data Science and Machine Learning Engineer

Remote

- Worked on freelance projects involving Data Science, Machine Learning, and Python development.
- Performed data preprocessing, exploratory data analysis, model training, and performance evaluation for client projects.
- Built machine learning solutions using Python, scikit-learn, Pandas, and related AI tools.

AWS Cloud Club - Zewail City

Dec 2025– Present

Technical Team Member

Giza, Egypt

- Collaborated with the technical team to organize and deliver workshops focusing on AWS cloud services, cloud architecture, and deployment strategies.
- Gained hands-on experience with core AWS services (e.g., EC2, S3, IAM) while assisting in building and managing cloud-based projects.
- Promoted cloud computing awareness across the university by mentoring peers, facilitating technical sessions, and supporting hands-on labs.

Technical Skills

Languages: Python, C++ (C++17), R, JavaScript, SQL

AI & Machine Learning: Computer Vision, Deep Learning, Explainable AI (XAI), Retrieval-Augmented Generation (RAG), Generative AI, Natural Language Processing (NLP), Symbolic AI, Information Retrieval, Statistical Inference

Libraries & Frameworks: PyTorch, Scikit-learn, Pandas, XGBoost, OpenCV, Flask, Streamlit, Qt Framework (PyQt), Selenium, Plotly Dash, NetworkX, Matplotlib, Imbearn, Optuna

Tools & Infrastructure: Docker, PostgreSQL, SQLite, CI/CD (GitHub Actions), FFmpeg, Pytest, Makefiles/qmake, SerpAPI, LaTeX

Projects

Tissue Insight Mega Project | Python, CNN, RAG, Machine Learning

2025

- Built a full-stack AI application for multi-modal cancer detection using gene expression analysis, computer vision, and Retrieval-Augmented Generation (RAG).
- Developed a tree-based classifier for gene expression analysis achieving a 91% F1-score and improving prediction accuracy by 35%.

- Implemented a CNN-based image classification pipeline to distinguish cancerous and non-cancerous microscopy samples.
- Integrated an interactive RAG chatbot to guide users through data upload and interpretation workflows.

Automatic Video Colorization Pipeline 🤖 | Python, PyTorch, ResNet34, U-Net, Computer Vision, FFmpeg **2025**

- Engineered a ResNet34-UNet encoder-decoder architecture to resolve the ill-posed problem of video colorization, regressing CIE Lab chrominance (ab) channels from grayscale luminance.
- Developed a category-aware data pipeline sampling 15,000 MS-COCO images, utilizing PyTorch AMP for mixed-precision training to achieve a 23.71 dB PSNR and 0.9235 SSIM.
- Implemented an end-to-end video inference engine using FFmpeg, applying Exponential Moving Average (EMA) temporal smoothing in chrominance space to eliminate inter-frame flicker.
- Successfully deployed the model to restore and colorize historic century-old public-domain footage, utilizing skip connections to preserve high-resolution spatial details.

Video Keyframe Summarization 🤖 | Python, PyTorch, Transformers, BiLSTM, Computer Vision, XAI **2026**

- Architected an end-to-end video summarization pipeline processing the TVSum dataset, computing deterministic temporal learning targets and caching frozen CNN visual embeddings for accelerated training.
- Engineered an Interpretable Transformer Encoder alongside a Compact BiLSTM baseline, extracting self-attention heatmaps to provide qualitative Explainable AI (XAI) evaluations of sequence selections.
- Implemented a unified inference engine applying a strict Top-15% temporal duration budget, generating comparative visualization masks to map prediction curves against ground-truth importance scores.
- Conducted cross-domain transfer learning evaluations on the SumMe dataset to validate model generalization, complemented by comprehensive architectural ablations comparing selection strategies.

Object Detection & State-Tracking Pipeline 🤖 | Python, PyTorch, YOLOv8, Computer Vision, OpenCV **2026**

- Engineered an end-to-end computer vision pipeline using YOLOv8 and PyTorch for real-time object detection and temporal state-tracking, achieving an mAP@50 of 98.19%.
- Developed a multi-layer state tracker utilizing aspect ratio, centroid drop, and proximity gating to accurately log temporal transitions and eliminate environmental false positives.
- Built a custom Python data-remediation module to automate video frame extraction, clamp malformed polygon coordinates, and curate a pristine ground-truth dataset.
- Documented the system architecture, evaluation metrics, and diagnostic plots in a comprehensive LaTeX scientific report and presentation.

Olivia Search Engine 🤖 | Python, Information Retrieval, BERT, BM25 **2025**

- Developed a modular search engine implementing foundational information retrieval techniques.
- Built an inverted index and Boolean query parser supporting AND/OR/NOT operations.
- Implemented TF-IDF and BM25 ranking algorithms for document retrieval and relevance scoring.
- Integrated ELMo and BERT embeddings for semantic retrieval, improving contextual search relevance.
- Improved retrieval accuracy by 12% using query expansion and semantic ranking techniques.

Network Traffic Multiclassification Model 🤖 | Python, XGBoost, Neural Networks **2025**

- Developed a machine learning system to classify network traffic into 6 behavioral categories.
- Preprocessed and normalized raw network traffic data while handling class imbalance and categorical encoding.
- Trained and evaluated Random Forest, XGBoost, and Neural Network models for anomaly detection and monitoring.
- Optimized performance using hyperparameter tuning and cross-validation techniques.

Adaptive Collaborative Code Learning Lab (ACCL) 🤖 | Python, Flask, Clean Architecture, SQLite, Pytest, Docker, CI/CD **2025**

- Collaborated as a core developer to engineer the majority of the backend infrastructure, implementing a strict Clean Architecture and MVC pattern using Python and Flask.
- Architected comprehensive domain services—including Authentication, automated Code Execution, and Plagiarism Detection—utilizing SQLAlchemy ORM and the Repository pattern.

- Integrated GenAI endpoints (Groq/Gemini) to provide automated, context-aware code hints for student submissions.
- Ensured backend reliability by developing a robust testing suite of over 560 unit and integration tests (74% coverage) using Pytest, orchestrated via GitHub Actions and Docker CI/CD pipelines.

ml-intrusion-detection-pipeline | Python, Scikit-learn, Imblearn, Optuna, SMOTE

2025

- Engineered a machine learning pipeline to detect network intrusions, utilizing **imblearn** and SMOTE to resolve severe class imbalance while strictly preventing cross-validation data leakage.
- Designed selective data preprocessing architectures, applying logarithmic scaling and standardization for distance-based models (SVM, KNN) while utilizing raw threshold logic for tree-based ensembles.
- Leveraged Optuna for Bayesian hyperparameter optimization, shifting SVM geometric boundaries (Linear to RBF) and optimizing Random Forest to achieve 99% precision and 96% recall.

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
Oracle University

Nov 2025 – Nov 2027
Credential ID: 103051607OCI25GAIOCP

Research

Uncibm: A Unified Library for Concept Bottleneck Models |

- Developing a comprehensive open-source framework designed to standardize and streamline the implementation of Concept Bottleneck Models (CBMs).
- Architecting the core library to facilitate highly interpretable deep learning pipelines, focusing on Explainable AI (XAI) methodologies.

Ongoing (Pre-publication)